Python Practice Questions with Answers

1. Create variables for: Name, Age, Phone number, Address. Display all.

```
name = "Binty" age =
25 phone =
"9876543210" address
= "Kerala"
 print("Name:", name)
print("Age:", age)
print("Phone:", phone)
print("Address:", address)
```

2. Store two numbers and find: Addition, Subtraction, Multiplication, Division.

```
a = 10 b = 5  print("Addition
=", a + b) print("Subtraction
=", a - b) print("Multiplication
=", a * b) print("Division =", a
/ b)
```

3. Swap two variables.

```
x = 20 y = 30
print("Before Swapping")
print("x =", x) print("y
=", y)
 temp =
x x = y
y = temp
print("After Swapping")
print("x =", x) print("y
=", y)
```

4. Find area of Circle.

```
radius = 7
circle_area = 3.14 * radius * radius

print("Area of Circle =", circle_area)
```

5. Find area of Rectangle.

```
length = 10 breadth = 5
rectangle_area = length * breadth
 print("Area of Rectangle =",
rectangle_area)
```

6. Find area of Triangle.

```
base = 8 height = 6 triangle_area =
0.5 * base * height
 print("Area of Triangle =",
triangle_area)
```

7. Calculate Simple Interest.

```
principal = 1000 rate = 5 time = 2
simple_interest = (principal * rate * time) / 100
 print("Simple Interest =",
simple_interest)
```

8. Calculate Compound Interest.

```
principal = 1000 rate = 5 time = 2  amount =
principal * (1 + rate / 100) ** time
compound_interest = amount - principal
 print("Compound Interest =",
compound_interest)
```

9. Convert Celsius to Fahrenheit.

```
celsius = 37
fahrenheit = (celsius * 9/5) + 32
 print("Fahrenheit =",
fahrenheit)
```

10. Convert Kilometers to Meters.

```
km = 5 meters = km
* 1000

print("Meters =", meters)
```

11. Convert Minutes to Hours.

```
minutes = 120 hours
= minutes / 60
 print("Hours =",
hours)
```

12. Take user input: Name, Age, College and display output.

```
user_name = input("Enter your name: ")
user_age = input("Enter your age: ") college
= input("Enter your college name: ")
 print("--- User Details ---
") print("Name:", user_name)
print("Age:", user_age)
print("College:", college)
```

Python Practice Questions and Answers Operators and Control Statements

1. Write a program to perform all arithmetic operations on two numbers.

```
num1 = float(input("Enter first number: ")) num2
= float(input("Enter second number: "))
 print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Modulus:", num1 % num2)
```

2. Check whether a number is odd or even using operators.

```
num = int(input("Enter a number: "))
 if num % 2 == 0:
print("Even")
else:
print("Odd")
```

3. Calculate simple interest using principal, rate, and time.

```
p = float(input("Enter principal amount: "))
r = float(input("Enter rate: ")) t =
float(input("Enter time: "))
 si = (p * r * t) /
100  print("Simple
Interest =", si)
```

4. Check whether a person is eligible to vote based on age.

```
age = int(input("Enter age: "))
 if age >=
18:
    print("Eligible to vote")
else:     print("Not
eligible")
```

5. Increase an employee's salary by 10% using assignment operators.

```
salary = float(input("Enter salary: "))
 salary += salary *
0.10
 print("Updated Salary =",
salary)
```

6. Check if a student passed all subjects using logical operators.

```
m1 = int(input("Enter mark1: ")) m2
= int(input("Enter mark2: ")) m3 =
int(input("Enter mark3: "))
 if m1 >= 40 and m2 >= 40 and m3 >=
40:
    print("Passed all subjects")
else:     print("Failed")
```

7. Find the largest among three numbers using comparison operators.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
 if a > b and a >
c:
    print("Largest =", a) elif
b > c:
    print("Largest =", b)
else:     print("Largest
=", c)
```

8. Create an ATM withdrawal system and update balance after withdrawal.

```
balance = 10000

withdraw = float(input("Enter withdrawal amount: "))

if withdraw <= balance:     balance -=
withdraw     print("Withdrawal
successful")     print("Remaining Balance
=", balance) else:
print("Insufficient balance")
```

9. Check whether a username exists in a list using membership operators.

```
users = ["binty", "rahul", "anu"]
 name = input("Enter username:
")
 if name in
users:
    print("Username exists")
else:     print("Username not
found")
```

10. Calculate BMI and determine the health category.

```
weight = float(input("Enter weight in kg: ")) height
= float(input("Enter height in meters: "))
```

```
 bmi = weight / (height **
2)
 print("BMI =",
bmi)
 if bmi <
18.5:
    print("Underweight") elif
bmi < 25:
    print("Normal") elif
bmi < 30:
    print("Overweight")
else:
print("Obese")
```

11. Create an electricity bill calculator based on units consumed.

```
units = int(input("Enter units consumed: "))
 if units <= 100:
bill = units * 5
elif units <= 200:
bill = units * 7
else:
    bill = units * 10

print("Electricity Bill =", bill)
```

12. Check loan eligibility using age, salary, and employment status.

```
age = int(input("Enter age: ")) salary =
float(input("Enter salary: ")) job =
input("Are you employed? (yes/no): ")
 if age >= 21 and salary >= 25000 and job ==
"yes":
    print("Eligible for loan")
else:     print("Not
eligible")
```

13. Compare two strings and determine whether they are equal.

```
str1 = input("Enter first string: ")
str2 = input("Enter second string: ")
 if str1 ==
str2:
    print("Strings are equal")
else:     print("Strings are not
equal")
```

14. Check whether a character exists in a given string.

```
text = input("Enter a string: ") char
= input("Enter a character: ")
 if char in
text:
    print("Character exists")
else:     print("Character not
found")
```

15. Compare two lists using == and is operators.

```
list1 = [1, 2, 3] list2
= [1, 2, 3]

print(list1 == list2) print(list1
is list2)
```

16. Calculate shopping discount based on purchase amount.

```
amount = float(input("Enter purchase amount: "))
 if amount >=
5000:
    discount = amount * 0.20 elif
amount >= 2000:
    discount = amount * 0.10 else:
    discount = 0

final_amount = amount - discount

print("Discount =", discount)
print("Final Amount =", final_amount)
```

17. Perform bitwise operations (&, |, ^, <<, >>) on two numbers.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
 print("AND =", a & b)
print("OR =", a | b)
print("XOR =", a ^ b)
print("Left Shift =", a << 1)
print("Right Shift =", a >> 1)
```

18. Update mobile balance after recharge using assignment operators.

```
balance = 100
 recharge = int(input("Enter recharge amount:
"))
 balance +=
recharge
```

```
 print("Updated Balance =",
balance)
```

19. Check employee bonus eligibility using experience and salary conditions.

```
experience = int(input("Enter years of experience: ")) salary
= float(input("Enter salary: "))
 if experience >= 5 and salary <
50000:
    print("Eligible for bonus")
else:     print("Not eligible")
```

20. Build a mini calculator to perform arithmetic operations based on user input.

```
num1 = float(input("Enter first number: ")) operator
= input("Enter operator (+, -, *, /): ") num2 =
float(input("Enter second number: "))
 if operator ==
"+":
    print("Result =", num1 + num2) elif
operator == "-":
    print("Result =", num1 - num2) elif
operator == "*":
    print("Result =", num1 * num2)
elif operator == "/":
    print("Result =", num1 / num2)
else:     print("Invalid
operator")
```

Conditional Statements – 10 Questions and Full Answers

1. Write a program to check whether a year is a leap year or not using conditional statements.

```
year = int(input("Enter year: "))
 if year % 400 == 0:
print("Leap Year")
elif year % 100 == 0:
    print("Not Leap Year")
elif year % 4 == 0:
print("Leap Year") else:
print("Not Leap Year")
```

2. Write a program to find the largest among 4 numbers entered by the user.

```
a = int(input("Enter first number: ")) b
= int(input("Enter second number: ")) c
```

```
= int(input("Enter third number: ")) d =
int(input("Enter fourth number: "))
 if a >= b and a >= c and a >=
d:
    print("Largest =", a) elif
b >= c and b >= d:
print("Largest =", b) elif c
>= d:
    print("Largest =", c)
else:     print("Largest
=", d)
```

3. Write a program to check whether a character is uppercase, lowercase, digit or special character.

```
ch = input("Enter a character: ")
 if ch >= 'A' and ch <=
'Z':
    print("Uppercase") elif
ch >= 'a' and ch <= 'z':
    print("Lowercase") elif
ch >= '0' and ch <= '9':
    print("Digit")
else:     print("Special
Character")
```

4. Write a program to calculate electricity bill.

```
units = int(input("Enter units: "))
 if units <= 100:
bill = units * 5
elif units <= 200:
    bill = (100 * 5) + ((units - 100) * 7) else:
    bill = (100 * 5) + (100 * 7) + ((units - 200) * 10)
 print("Bill =",
bill)
```

5. Write a program to check whether three sides can form a triangle and find the type.

```
a = int(input("Enter side1: ")) b
= int(input("Enter side2: ")) c =
int(input("Enter side3: "))
 if a + b > c and a + c > b and b + c >
a:     if a == b and b == c:
        print("Equilateral Triangle")
elif a == b or b == c or a == c:
print("Isosceles Triangle")     else:
        print("Scalene Triangle")
else:     print("Not a Triangle")
```

6. Write a program to input marks of 5 subjects and print grade.

```
m1 = int(input("Enter mark1: ")) m2
= int(input("Enter mark2: ")) m3 =
int(input("Enter mark3: ")) m4 =
int(input("Enter mark4: ")) m5 =
int(input("Enter mark5: "))

total = m1 + m2 + m3 + m4 + m5 avg
= total / 5
 if m1 < 50 or m2 < 50 or m3 < 50 or m4 < 50 or m5 <
50:
    print("Fail") elif
avg > 90:
    print("Grade A") elif
avg >= 75:
    print("Grade B") elif
avg >= 50:
print("Grade C") else:
print("Fail")
```

7. Write a program to check whether a number is positive even, positive odd, negative even, negative odd or zero.

```
num = int(input("Enter number: "))
 if num ==
0:
    print("Zero")
elif num > 0:     if
num % 2 == 0:
        print("Positive Even")
else:
        print("Positive Odd")
else:     if num % 2 == 0:
        print("Negative Even")
else:         print("Negative
Odd")
```

8. Write a program to create a simple login system.

```
username = input("Enter username: ")
password = input("Enter password: ")
 if username == "admin":
if password == "1234":
        print("Login Successful")
else:
        print("Incorrect Password")
else:     print("User Not Found")
```

9. Write a program to check eligibility for voting, driving license and senior citizen discount.

```
age = int(input("Enter age: "))
 if age >=
18:
    print("Eligible for voting") else:
    print("Not eligible for voting")
 if age >=
18:
    print("Eligible for driving license")
else:
    print("Not eligible for driving license")
 if age >=
60:
print("Eligib
le for senior
citizen
discount")
else:
print("Not
eligible for
senior
citizen
discount")
```

10. Write a program for a mini ATM system.

```
pin = int(input("Enter PIN: "))
balance = 5000  if pin == 1234:
    print("PIN Correct")     amount =
int(input("Enter withdrawal amount: "))
     if amount <=
balance:
        balance = balance - amount
print("Withdrawal successful")
print("Balance =", balance)     else:
        print("Insufficient Balance")
else:     print("Wrong PIN")
```

Python Programs – For Loop and While Loop

1. Print numbers from 10 to 1

```
num = 10
for i in range(1, num + 1):
    print(num)
    num -= 1
```

2. Print odd numbers from 1 to 50

```
print("Odd numbers between 1 and 50")
for i in range(1, 50, 2):
    print(i)
```

3. Find the sum of numbers from 1 to 100

```
num = 100
sum = 0
for i in range(1, num + 1):
    sum += i
print("sum of numbers from 1 to 100", sum)
```

4. Print the multiplication table of any number entered by the user

```
num1 = int(input("Enter any number "))
for i in range(1, 11):
    print(num1, "x", i, "=", num1 * i)
```

5. Count how many digits are present in a number using a while loop

```
num = int(input("Enter a number: "))

count = 0

while num > 0:
    num = num // 10
    count += 1

print("Number of digits =", count)
```